Q1. Reference Resolution, Discourse

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
ORIGINAL DISCOURSE:
Ravi met Arun at the library. He borrowed a book and later returned it.

RESOLVED DISCOURSE:
Ravi met Arun at the library. Ravi borrowed a book and later returned book.

COREFERENCE CHAINS:
Ravi -> Ravi -> He
Arun -> Arun
library -> library
book -> book -> it
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

Q2. Text Coherence, Discourse Structure

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
TEXT COHERENCE ANALYSIS
--------------------------------------------------
Sentence 1: The roads were flooded after heavy rainfall.
Sentence 2: Therefore, schools were closed for the day.
Relation: Cause-Effect

Sentence 1: Therefore, schools were closed for the day.
Sentence 2: Students attended classes online.
Relation: Sequence

PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

Q3. Dialog Acts, Conversational Agents

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
DIALOGUE ACT RECOGNITION
--------------------------------------------------
Speaker: User
Utterance: Can you book a train ticket for me?
Dialogue Act: Request

Speaker: Agent
Utterance: Sure, where would you like to travel?
Dialogue Act: Question

Speaker: User
Utterance: I want to go to Chennai.
Dialogue Act: Inform

Speaker: Agent
Utterance: Your ticket has been booked.
Dialogue Act: Confirmation / Action

DIALOGUE ACT SEQUENCE:
Request -> Question -> Inform -> Confirmation / Action
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

Q4. Language Generation, Surface Realization

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\p
ython.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
SEMANTIC REPRESENTATION:
Action: Buy
Agent: Student
Object: Book
Tense: Past

GENERATED SENTENCE:
The student bought a book.
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

Q5. Machine Translation, Interlingua, Statistical Approaches

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\p
ython.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
event: PLAY
agent: BOY
theme: FOOTBALL
tense: PRESENT
aspect: PROGRESSIVE

TRANSLATION CANDIDATES:
लड़का फुटबॉल खेल रहा है।
LM Score: 0.95
TM Score: 0.96
Combined Score: 0.955

फुटबॉल लड़का खेल रहा है।
LM Score: 0.4
TM Score: 0.7
Combined Score: 0.55

लड़का खेल रहा है फुटबॉल।
LM Score: 0.3
TM Score: 0.5
Combined Score: 0.4

FINAL TRANSLATION:
लड़का फुटबॉल खेल रहा है।
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```